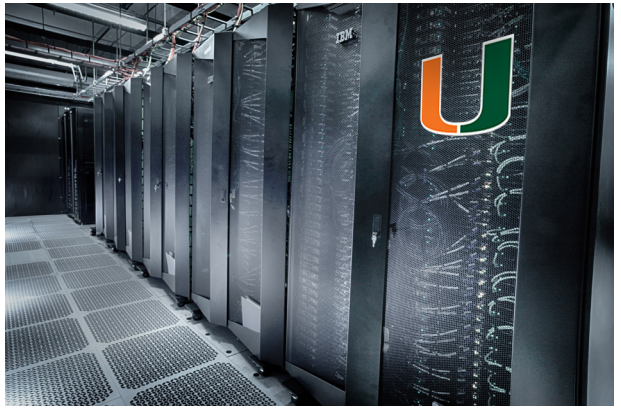
Getting started with python on Pegasus

Ray Bell

Lunch Bytes. Fall 2018



$ git clone https://github.com/milancurcic/lunch-bytes.git

$ cd lunch-bytes/Fall_2018/LB14

Installing python on pegasus

- Setup folders:

$ mkpir -p src local/bin

$ cd src

- Install miniconda into your directory https://conda.io/miniconda.html

$ wget https://repo.continuum.io/miniconda/Miniconda3-latest-Linux-x86_64.sh

$ chomd +x Miniconda3-latest-Linux-x86_64.sh

$ ./Miniconda3-latest-Linux-x86_64.sh [-u]

Install in /nethome/USERNAME/local/bin/miniconda3

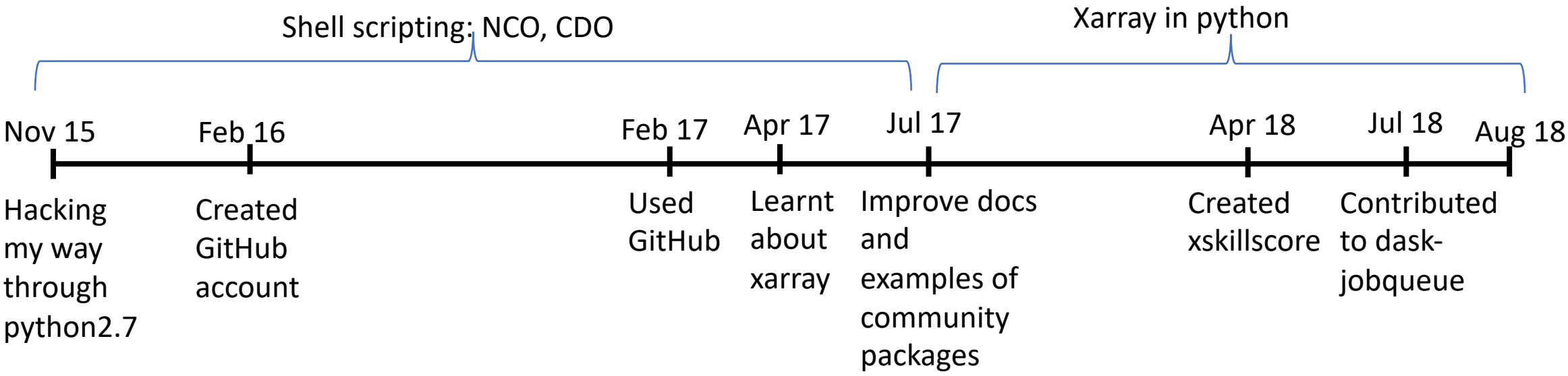
Creating an environment file

On laptop
$ conda env create -f py_intro_lb_env_file.yml
$ conda activate py_intro_lb_env

On Pegasus (can take up to 20 mins…)
$ bsub < submit_create_env.sh
$ bjobs
$ bpeek
$ conda info --envs # after job finished. Should see py_intro_lb_env
$ conda activate py_intro_lb_env
[Note: when I tested this morn dask-jobqueue didn’t install
$ pip install git+https://github.com/dask/dask-jobqueue]

About me

- Started as a post-doc in November 2015. Previously used IDL and Matlab.
- Had a couple of options: NCL or Python.
- Leo Siqueira used python and could help me!



Pegasus

- See seminar by Natalie Perlin on September 14th

What is python?



- Created by Guido Van Rossum in 1991
- Open source (free!)
- Object orientated (procedures can access and modify them ‘self’). E.g.

Rolling seasonal average of monthly data:

```
pd.DataFrame({'A': [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]}).rolling(3).mean()
```

Python package (pandas).
Import pandas as pd

How to create data/array in pandas

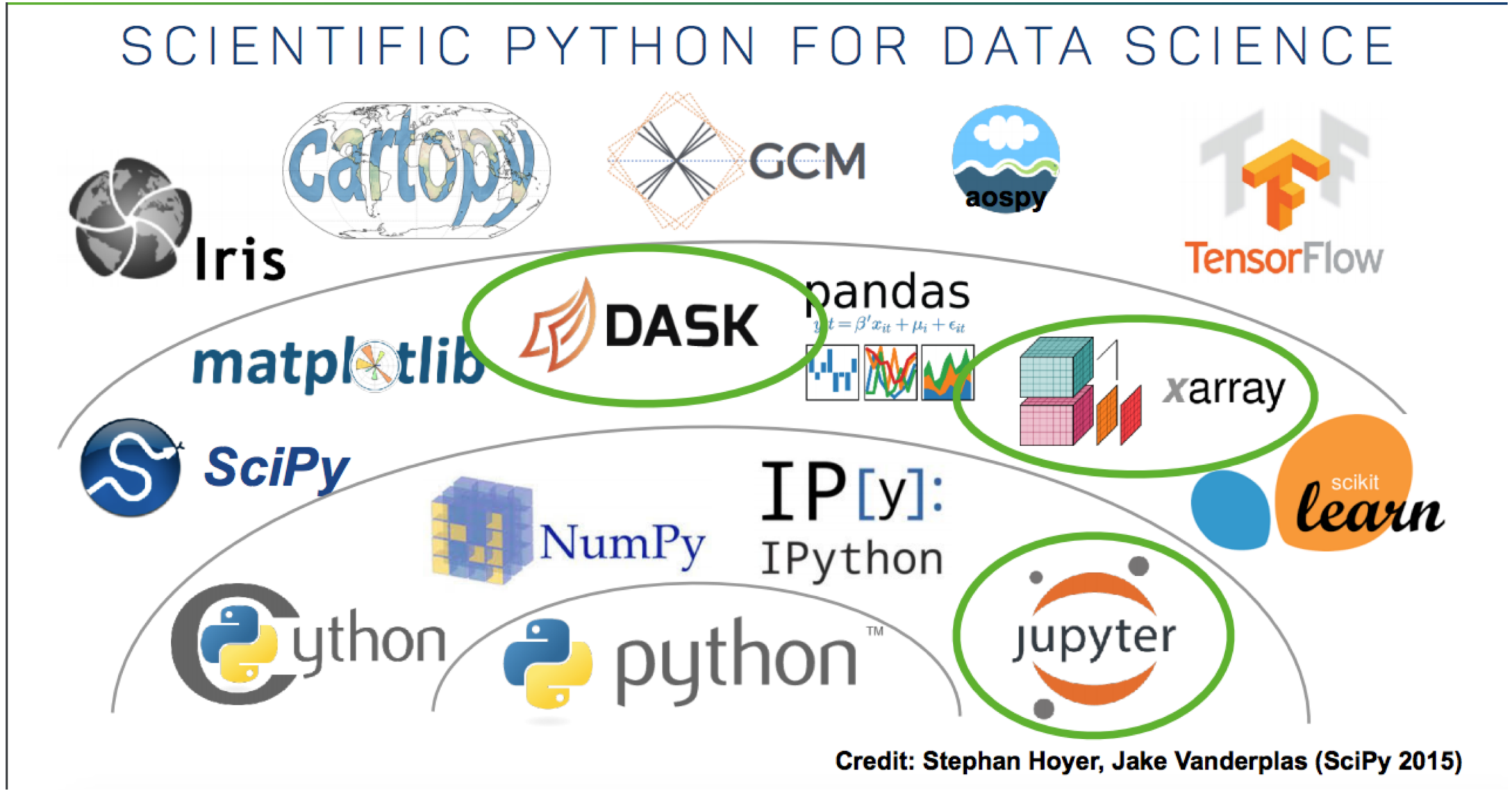
Rolling operator over 3 values

Take the mean

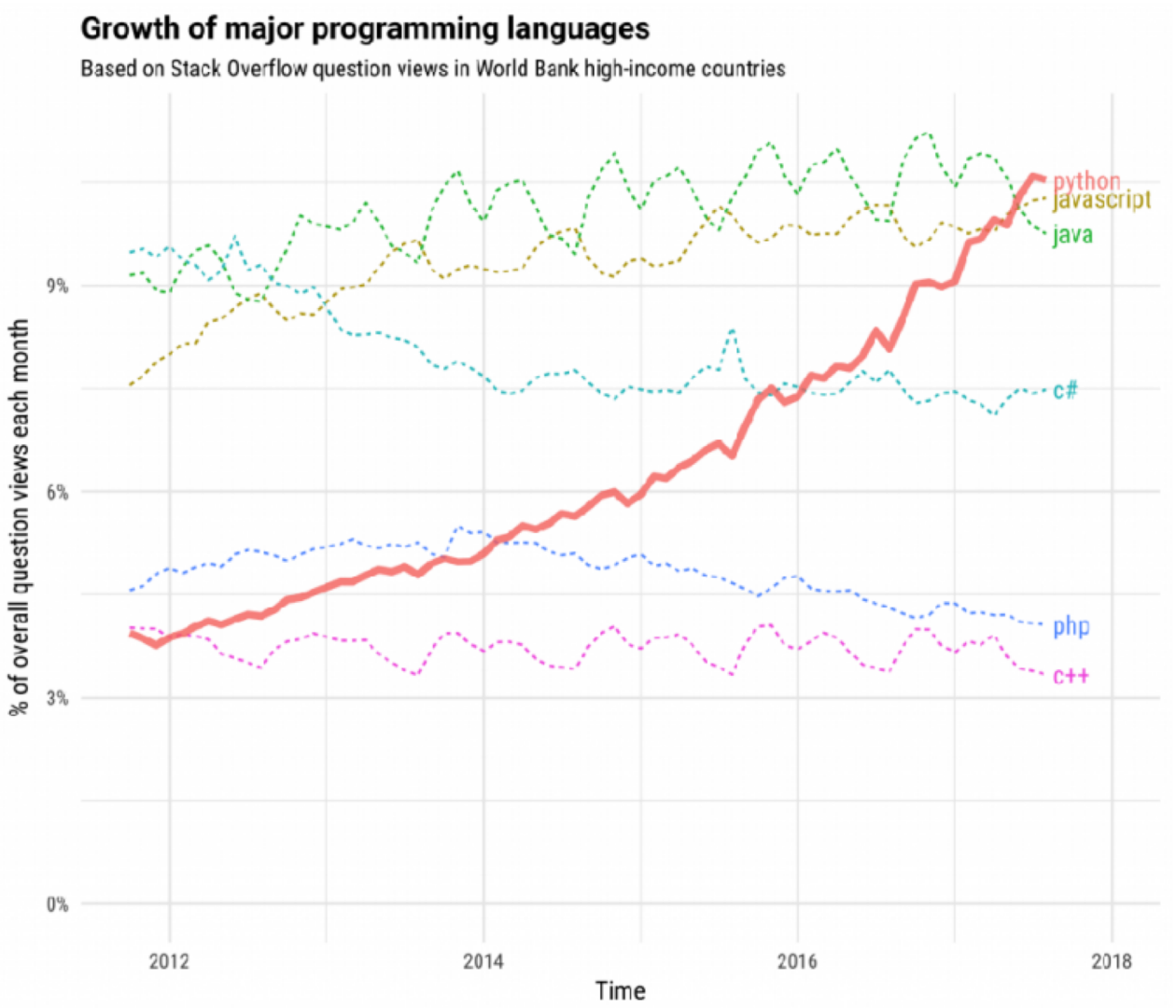
What is python?

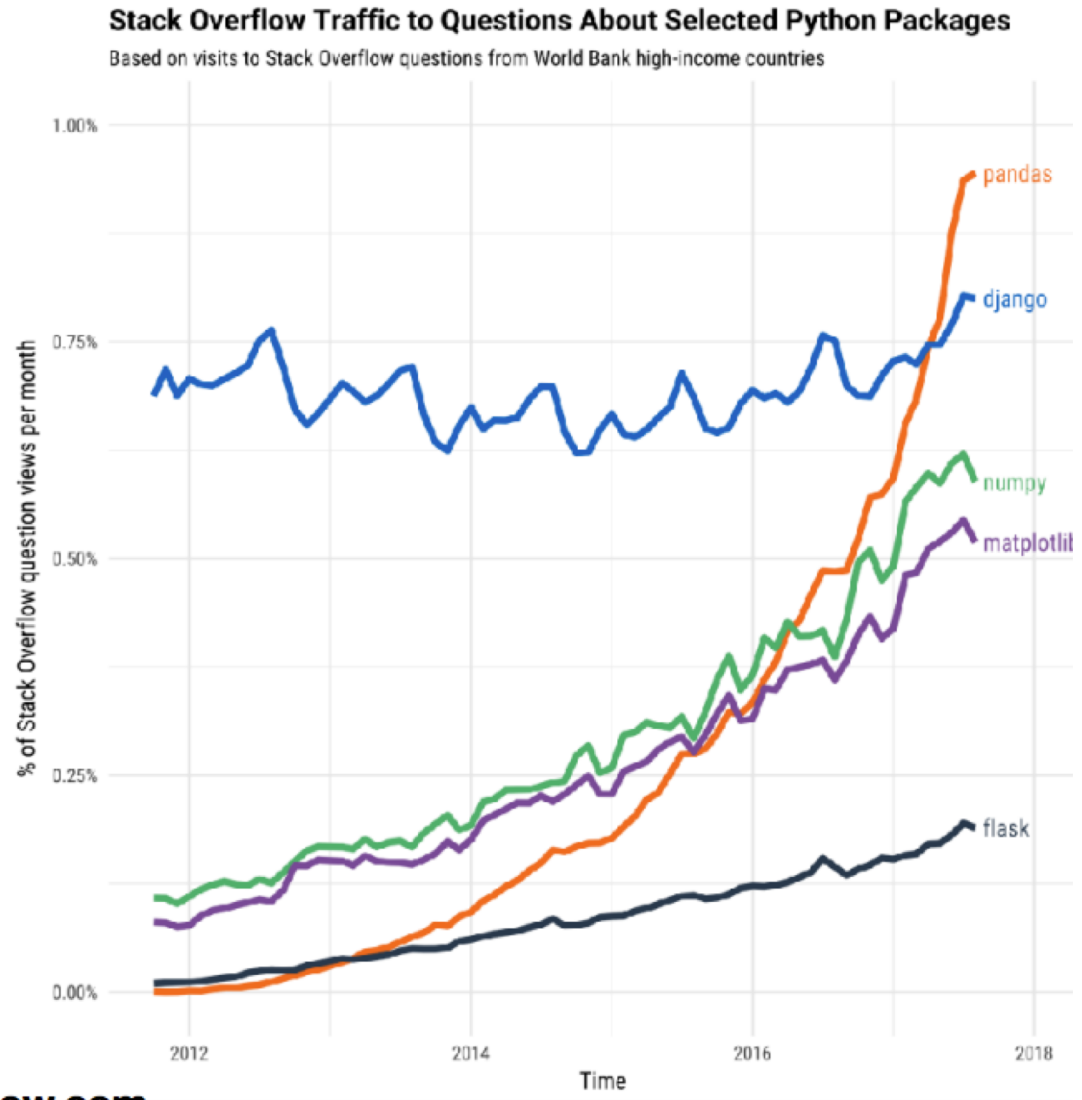
- Funding from DARPA (Defense Advanced Research Projects Agency) in ~2000. Proposal entitled ‘Computer Programming for Everybody’ https://www.python.org/doc/essays/cp4e/
- Python Software Foundation ‘board members’ and community effort work on improving python. https://devguide.python.org/
- Agree on PEPs (Python Enhancement Proposals) https://www.python.org/dev/peps/
- NumFOCUS provide support for some packages e.g. numpy. Matplotlib
- PyCon, SciPy, pyladies

What is python?



Why python?





source: stackoverflow.com

Why python

- Large uptake by the geoscientific community e.g. unidata workshops (https://www.unidata.ucar.edu/events/) and ECMWF conference (https://www.ecmwf.int/en/learning/workshops/2018-workshop-developing-python-frameworks-earth-system-sciences)
- Easy to stay on top of developing packages.
- ~150,000 packages (https://pypi.org/)
- Industry applications.

Python resources

- RSMAS python list serv. Send an e-mail to python-subscribe@lists.rsmas.miami.edu
- Stackoverflow https://stackoverflow.com/questions/tagged/python and how to ask questions see https://stackoverflow.com/help/mcve
- GitHub
- Python people at RSMAS: Milan, Leo, Tiago, Suvy, Brian

- Text editors: https://atom.io/ https://pythonhosted.org/spyder/editor.html

Overview

- Install miniconda
- Python
- ipython
- SSH tunneling for jupyter notebooks
- Dask-jobqueue for parallel jobs

Python strings

http://swcarpentry.github.io/python-novice-inflammation/
Other resources?

$ python
>>> print('Hello RSMAS!’)
>>> word = ‘lead’

Indexing starts at 0
>>> word[0]

What happens when you type >>> word[4]?

Python for loops

Use 4 spaces for indentation:

```
>>> for char in word:
...     print(char)
```

Create a sequence of numbers

```
>>> for i in range([start],4):
...     print(i)
```

Python basic math

```
>>> 1 + 2
>>> 1 / 2
>>> 2 * 3
>>> 2 ** 3
```

Python scripts

Add to mypyscript.py:

```
mylist = [10, 7, 5, 18]
for i, val in enumerate(mylist):
    print(i, val)
```

```
$ python mypyscript.py
```

Python functions

Add to func_script.py:

```
def fib(n):
    """ Return Fibonacci series up to n """
    result = []
    a, b = 0, 1
    while a < n:
        result.append(a)
        a, b = b, a + b
    return result
print(fib(10))
```

$ python func_script.py

Or

```
$ python
>>> import func_script as fs
>>> fs.fib(10)
```

ipython

- Interactive python. Works best on a notebook on laptop.
- Has useful ‘magic words’ https://ipython.readthedocs.io/en/stable/interactive/magics.html

$ ipython

In [1]: %time sum(range(1000000))

In [1]: alpha = 123

In [2]: beta = 'test’

In [3]: %whos

In [1]: %run myscript.py

ipython

Add to buggy_script.py:

```
mylist = [10, 7, 5, 18]
for i, val in enumerate(mylist):
    print(i, v)
```

%run buggy_script.py

https://docs.python.org/3.6/tutorial/errors.html

%debug [‘u’ for up Traceback and ‘d’ for down Traceback]

environments

- How not to fudge up your work space.
- https://conda.io/docs/user-guide/tasks/manage-environments.html

$ conda create --name test_env python=3.6

[$ conda remove --name test_env --all]

[$ conda info --envs]

$ conda activate test_env

Installing packages

- Quick way: $ pip install PACKAGE
- Get the bleeding edge version: $ pip install git+https://github.com/pydata/xarray.git

- Recommended to use conda / conda forge https://conda-forge.org/

$ conda install -c conda-forge notebook

Plotting

Add to plotting_script.py

```
import matplotlib.pyplot as plt
plt.plot([1,2,3,4])
plt.savefig('test.png')
```

```
$ python plotting_script.py
```

Jupyter notebook for plotting

- See seminar by Suvy next week

Configuring the notebook

- http://pangeo.io/setup_guides/hpc.html#configure-jupyter

$ jupyter notebook --generate-config

This creates ~/.jupyter/jupyter_notebook_config.py

$ ipython

In [1]: **from notebook.auth import** passwd; passwd()

Copy the password hash to the c.NotebookApp.password line in ~/.jupyter/jupyter_notebook_config.py

$ chmod 400 ~/.jupyter/jupyter_notebook_config.py

Tunneling the notebook

$ jupyter notebook --no-browser --ip=10.141.226.234 --port=8888
[$ jupyter notebook --no-browser]

On my laptop open a terminal and do
$ ssh -N -L 8888:10.141.226.234:8888 USERNAME@pegasus.ccs.miami.edu
[$ ssh –N –L 8888:localhost:PORTNUMBER USERNAME@pegasus.ccs.miami.edu]

Open http://localhost:8888 on laptop

New -> Notebook: Python 3

Jupyter notebook for plotting

- See https://github.com/milancurcic/lunch-bytes/blob/master/py_intro/py_intro_lb_nb_plotting.ipynb

The examples use:

https://matplotlib.org/

https://pandas.pydata.org/pandas-docs/stable/visualization.html

https://seaborn.pydata.org/

https://scitools.org.uk/cartopy/docs/latest/

http://xarray.pydata.org/en/stable/plotting.html

MyBinder

- See https://github.com/raybellwaves/SSElFaro

https://mybinder.org/

Packages for interactive plotting:
https://bokeh.pydata.org/en/latest/
http://geo.holoviews.org/
https://altair-viz.github.io/
https://ipyleaflet.readthedocs.io/en/latest/
http://python-visualization.github.io/folium/

Submitting parallel jobs using dask-jobqueue

- See https://github.com/milancurcic/lunch-bytes/blob/master/py_intro/py_intro_lb_nb_parallel.ipynb

https://dask-jobqueue.readthedocs.io/en/latest/

https://distributed.readthedocs.io/en/latest/

To find the IP address to tunnel

```
$ python
>>> from dask_jobqueue import LSFCluster
>>> from dask.distributed import Client
>>> cluster = LSFCluster(cores=2, memory='4 GB’)
>>> client = Client(cluster)
>>> print(client)
```